

United States — Substation ESG Report ^{v4.2}

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks, across the 101,594-substation United States fleet. Updated monthly via the SSI Index v4.2 scoring engine (101 variables · 34 sources · 11 modifiers · Re composite bounded [0.920, 1.787]).

SUBSTATION

Fermilab Kautz Road Substation

120000037277 · Illinois, Illinois

SSI SCORE (R_{MEDIAN})

0.895

● Extreme Risk

87.0th percentile

VOLTAGE CLASS

345 kV

High-voltage transmission

CONFIDENCE INTERVAL

0.300 – 0.300

P5–P95 · CI width 0.240 · high confidence

MARKOV ETTC

14.7 yr

Expected time to criticality

DER PENETRATION

0.60

T1 transition stress score

Component Fingerprint ^{+ Re composite v4.2}

Six-component weighted decomposition of the SSI composite score (C / V / I / E / S / T) — the substation's structural risk profile — PLUS the v4.2 **Re composite**, a bounded resilience aggregate of the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just). Layout: *left* — the 6 components rendered as a weighted contribution bar · *right* — the same 6 components rendered as horizontal bars side-by-side with Re_norm as the comparable 7th bar (terracotta) on the [0, 1] scale · *below* — the Re composite tier card (numeric value + bounded trajectory bar + classification + formula explainer). Re composite re-appears as the 7th axis of the ESG Radar in Section A below.

WEIGHTED CONTRIBUTION TO R_{BASE}

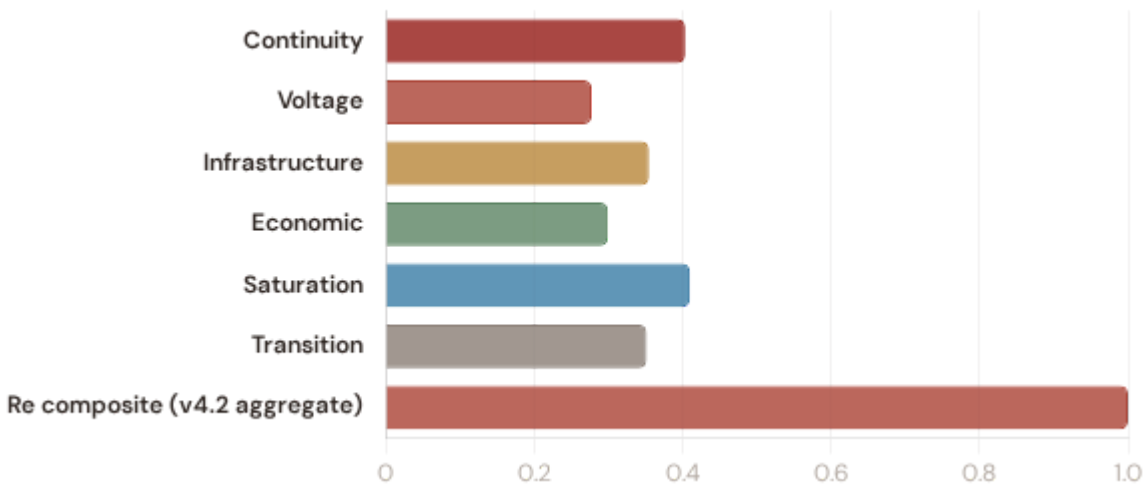


V4.2 RE COMPOSITE (AGGREGATE, NOT IN R_{BASE}) elevated exposure



Continuity 0.404 × 0.3 Voltage 0.278 × 0.1 Infrastructure 0.355 × 0.25 Economic 0.300 × 0.1
Saturation 0.410 × 0.2 Transition 0.352 × 0.05

R_{base} total (weighted sum) 0.367 $\Sigma = 1.00$



1.000

RE_NORM

ELEVATED EXPOSURE

LOW MODERATE ELEVATED (1.00)
(0.00) (0.25) (0.50)



Re_raw 1.7870 (bounds [0.920, 1.787]; normalised to [0, 1])

FORMULA

$$Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$$

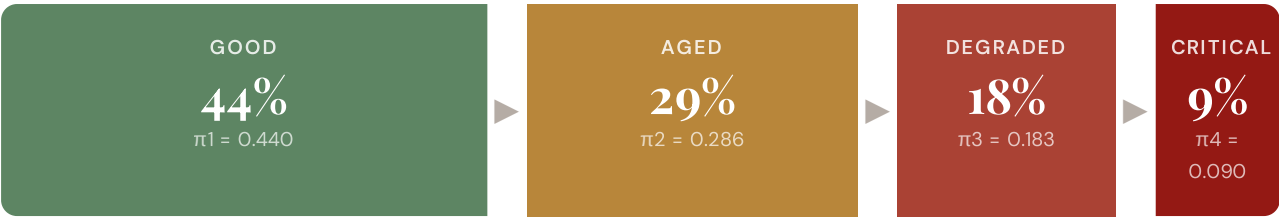
Bounded resilience aggregate of the v4.2 family. $Re < 1.0$ when R8 adaptive capacity dominates other near-neutral factors.

Re composite tier classification: **low** ($Re_norm < 0.25$) · **moderate** ($0.25-0.50$) · **elevated** (≥ 0.50). Bounds [0.920, 1.787] map to Re_norm [0, 1] via $\text{clip}((Re_raw - 0.920) / (1.787 - 0.920), 0, 1)$.

Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

STEADY-STATE DISTRIBUTION



Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.

RISK SCORE

0.207

Composite Markov risk

ETTC

14.7 yr

Expected time to critical

P(CRITICAL) 20YR

100.0%

Probability of reaching critical

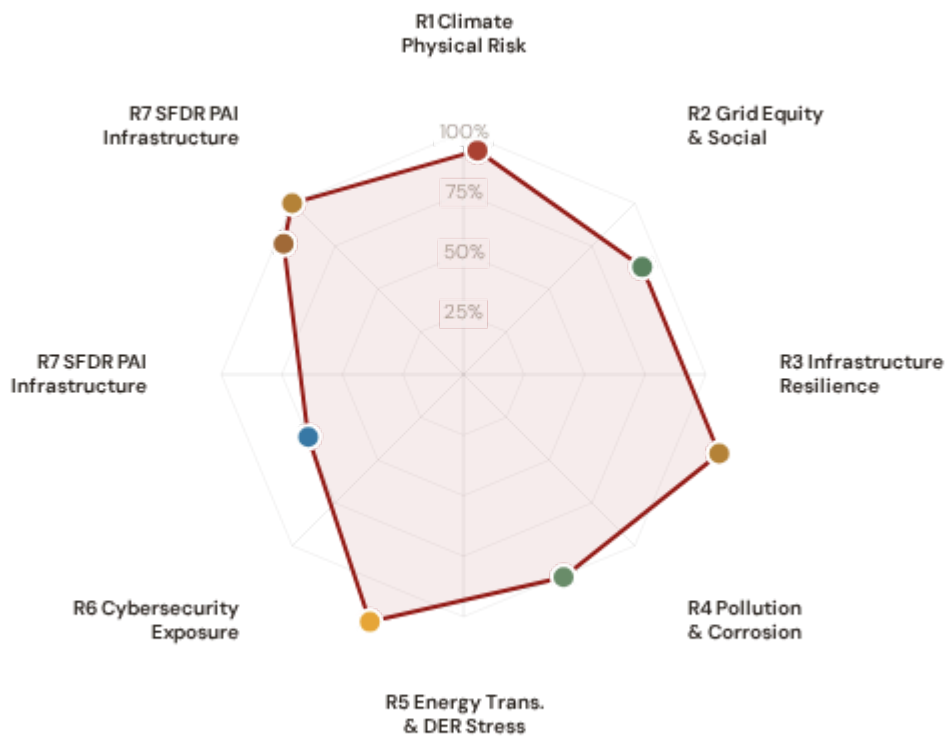
CORROSION

C3

ISO 9223 classification

ESG Radar — Seven-Report Overview ^{v4.2}

A Composite readiness across 7 ESG dimensions per FC v3 §14 (R1 Climate Physical · R2 Grid Equity & Social · R3 Infrastructure Resilience [Re composite ESG home] · R4 Pollution & Corrosion · R5 Energy Transition & DER Stress · R6 Cybersecurity · R7 SFDR PAI Infrastructure, FC v3 §14 subsection 13.7), each linked to UN Sustainable Development Goals. *Note: R7 here is the SFDR ESG-disclosure axis (Re_norm proxy under Convention #7) — distinct from the v4.0.2 formula-chain R7 cyber/digital modifier documented in methodology.html. The two R7s share a number by design (V4_2_IMPLEMENTATION_ARCHITECTURE.md §4.5).*



Re composite v4.2 7th axis · Re_raw 1.7870 · Re_norm 1.000 · ELEVATED EXPOSURE



R1 · Climate Physical Risk Assessment READY

SDG 13 Climate Action · SDG 9 · SDG 11

SDG 13

SDG 9

SDG 11

R2 · Grid Equity & Social Vulnerability READY

SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1

SDG 7

SDG 10

SDG 1

R3 · Infrastructure Resilience [Re composite home] READY

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12

SDG 9

SDG 13

SDG 12

R4 · Pollution & Corrosion READY

SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15

SDG 11

SDG 3

SDG 15

R5 · Energy Transition & DER Stress READY

SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9

SDG 7

SDG 13

SDG 9

R6 · Cybersecurity Exposure PARTIAL

SDG 9 Industry, Innovation, Infrastructure · SDG 16

SDG 9

SDG 16

R7 · SFDR PAI Infrastructure Disclosure READY

SDG 12 Responsible Consumption and Production · SDG 9 · SDG 13

SDG 12

SDG 9

SDG 13

R7 · SFDR PAI Infrastructure GAP v4.2

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 11 · Re_norm = 1.000 (elevated exposure)

SDG 9

SDG 13

SDG 11

1 Climate Physical Risk Assessment

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

READY

WHY THIS REPORT EXISTS

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate change adaptation activities to demonstrate that physical climate risks have been identified and addressed. This report quantifies acute and chronic climate hazard exposure at substation level using ERA5 reanalysis and Markov degradation modelling.

SDG Alignment**SDG 13 — Climate Action** (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI IRI metrics (snow/ice, wind, heat stress) and Markov degradation model directly quantify how climate hazards affect this substation's operational resilience over 10- and 20-year horizons. With a Markov risk score of 0.207 and ETTC of 14.7 years, this substation's climate trajectory can be assessed against TCFD physical risk and EU Taxonomy adaptation screening requirements.

SDG 13

SDG 9

SDG 11

Substation Profile

Insulation Risk (I component)	0.355
Environment Component (E)	0.300
Seismic PGA	0.0259g (Zone 4)
Markov Risk Score	0.207
ETTC (time to criticality)	14.7 years
P(critical) at 20 years	100.0%
Corrosion Class	C3
Climate Trajectory (R2)	100.0% (20yr Markov)

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
I1 Snow/Ice IRI	0.983	ESRS E1-9: Financial effects from physical risks	READY
I2 Tree-fall IRI	1.050	TCFD Strategy (b): Physical risk impact	READY
I3 Heat-wave IRI	1.026	EU Taxonomy Annex A: Heat stress	READY
I5 Thermal stress (I aggregate)	0.355	ESRS E1-4: Climate mitigation targets	DEFAULT
R2 Climate trajectory	100.0% (20yr)	TCFD Strategy (c): Scenario analysis	READY
R6b Seismic PGA	0.0259g	EU Taxonomy Annex A: Geophysical hazards	READY
Markov p_critical_10yr	100.0%	TCFD Risk Management: Risk identification	READY
Markov p_critical_20yr	100.0%	ESRS E1-9: Time horizons	READY

Data Sources & Vintage

ERA5 Climate Reanalysis

Copernicus CDS

2024 Weekly

National Seismic Hazard Map

National Geological Survey

2023 Multi-year

IEEE C57.91 Thermal Model

IEEE

Standard N/A

CIGRE TB 761 Markov

CIGRE

2019 N/A

CMIP6 SSP2-4.5 Projections

Copernicus CDS

2024 v4.2 5-model ensemble

TECHNICAL VALIDITY

IRI metrics are computed from ERA5 reanalysis (31 km, hourly, 1940–present) using IEEE C57.91 thermal loading curves. Markov 5-state degradation is calibrated from CIGRE Technical Brochure 761 (2019) condition assessment data. Forward-looking projections use the CMIP6 SSP2-4.5 5-model ensemble (ACCESS-CM2, CNRM-CM6-1, EC-Earth3, GFDL-ESM4, MRI-ESM2-O), baseline 2000–2020 vs future 2030–2050, feeding the substation-level climate_trajectory (I1/I2/I3) fields for 32 of 39 countries. Any PARTIAL status now reflects country-specific gaps: 7 Wave 2–3 countries (Australia, Chile, Colombia, Costa Rica, Iceland, Ireland, Israel) await the next L2 climate pass; 8 Wave 4 countries emit fleet-uniform trajectories pending the per-substation bilinear interpolation regression fix (methodology follow-on, sister to R3_C_mult uniformity). All other inputs are production-grade with institutional provenance.

2 Grid Equity & Social Vulnerability

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

READY

WHY THIS REPORT EXISTS

ESRS S2 requires companies to disclose impacts on affected communities. SFDR mandates 5 social PAI indicators. The SSI Index is uniquely positioned here because no competing grid risk framework integrates socio-economic vulnerability at substation resolution. This report surfaces where infrastructure risk concentrates in economically disadvantaged populations.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. This substation serves a catchment with V_socio of 0.18 and energy poverty rate of 10%. The R3 social modifier (0.973) amplifies risk scores in regions with high unemployment, elderly concentration, and net outward migration — making spatial inequality visible and quantifiable at asset level.

SDG 7

SDG 10

SDG 1

Substation Profile

V_socio (energy poverty index)	0.18
EP Rate (energy poverty %)	10%
Unemployment Rate	5.9%
GDP per Capita	€ 78,433
R&D % of GDP	3.7%
R3 Social Multiplier	0.973
Elderly Vulnerability	16.23%
Migration Score	0.98

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
V_socio Energy poverty index	0.18	ESRS S2-4: Impacts on affected communities	READY
EP_rate Energy poverty rate	10%	SFDR PAI #14: Fossil fuel exposure proxy	READY
R3 Social multiplier	0.973	ESRS S1-6: Workforce characteristics	READY
Elderly Elderly vulnerability	16.23%	SFDR PAI: Vulnerable populations	READY
Community Catchment population	75,101	ESRS S2: Community impact	READY

Data Sources & Vintage

GHSL Population Grid EC JRC / Copernicus	2025 Multi-year
Global Net-Migration Grid Niva et al. 2023 (Aalto / Wittgenstein)	2000-2019 Multi-year
Eurostat GISCO NUTS-3 Regions European Commission / Eurostat	2024 Annual
GADM Administrative Areas 4.1 (CO / CH / CA) UC Davis geodata.ucdavis.edu/gadm	2022 Multi-year
Economic Statistics (GDP, unemployment) National Statistics Office	2023 Annual
Energy Market Data National TSO	2023 Annual

TECHNICAL VALIDITY

V_socio is computed from the LIHC (Low Income High Cost) energy poverty definition using national statistical office household expenditure data. The R3 modifier amplifies infrastructure risk scores in catchments with high unemployment, elderly concentration, and net outward migration. Coverage status (21 July 2026 audit): V_socio + EP_rate + gdp_per_capita + unemployment_rate are populated for 32/39 countries, but 8 large countries (Italy, Japan, Portugal, Spain, Sweden, Germany, France, US) emit fleet-uniform national aggregates instead of per-substation LAU-2/NUTS-3 values — pending the systemic Wave 4 per-substation interpolation regression fix (sister to R3_C_mult + CMIP6 climate_trajectory). (Task #452 already closed this gap for migration_score specifically via per-substation Niva 2023 raster enrichment; the remaining V_socio / EP_rate / GDP / unemployment fields await Task #450 SYSTEMIC scope.) Catchment

population is now populated for all 39/39 countries via the GHSL Population Grid (EC JRC / Copernicus Emergency Management Service, R2023A epoch E2025, Mollweide equal-area) computing 5 km per-substation zonal sums — Task #451 landed 23 July 2026, replacing a legacy synthetic generator with a real GHSL spatial-join pipeline (795k-substation cohort; 0.17% receive Convention #56 visibly-honest None fallbacks for remote/offshore points where the raster has NoData — concentrated in US Aleutians+Pacific, Canada Arctic, Australia outback, Nordic fjords). Migration_score is now populated for all 39/39 countries via the Niva et al. 2023 gridded net-migration dataset (Aalto University / Wittgenstein Centre, published Nature Human Behaviour 7:2023–2037, DOI 10.1038/s41562-023-01689-4, dataset DOI 10.5281/zenodo.7997134, CC BY 4.0 licence) using the 20-yr sum raster 2000–2019 at 5 arc-min (~10 km) resolution — Task #452 landed 23 July 2026, closing the R2 defect via NARROW-scope enrichment (fill Nones) plus fleet-uniform override that retired the 0.5 national scalar fallback of 8 Wave 4 majors (France, Germany, US, Italy, Spain, Portugal, Sweden, Japan, Australia partial, Austria partial) with real per-substation Niva-derived values in [0, 1] range via $\tanh(x/200)$ mapping. Genuine pre-Task-#452 per-substation distributions (Norway, Denmark, Lithuania, Ireland, Turkey, Mexico, Greece, New Zealand, Poland partial) are preserved untouched — their semantic-drift issues (e.g. Mexico [-5, +8] percent-scale, Denmark tiny [-0.02, +0.04] scale) are deferred to Task #450 SYSTEMIC scope. Cohort-wide 97.13% real Niva values written + 0.17% Convention #56 visibly-honest None + 2.70% preserved distributions untouched. The remaining 4 countries with major partial-coverage gaps from failed spatial joins (Luxembourg 12% populated / Slovenia 16% / Colombia 51% / Lithuania 50%) are now closed via Task #453 (23 July 2026) using per-substation polygon spatial-join against Eurostat GISCO NUTS-3 2024 shapefile (European Commission / Eurostat, EPSG:4326, annual vintage, open-license attribution-required — for LU/SI/LT) plus GADM 4.1 admin1 shapefile for Colombia (UC Davis geodata.ucdavis.edu/gadm, peer-reviewed academic-open-license derivative of DANE Marco Geoestadístico Nacional — chosen as documented-proxy after DANE geoportal empirically requires form-based interactive download). Task #453 empirical closure: 6,967 v43 substations enriched across LU/SI/LT/CO (LU 634/634 = 100% · SI 1,571/1,574 = 99.8% · LT 4,396/4,396 = 100% · CO 366/366 = 100%; 3 Convention #56 fallback total = 0.04% cohort-wide; 100% preservation of Task #451 catchment_population + Task #452 migration_score audit-trail markers via merge-not-replace BINDING contract). Each Task #453 enriched substation carries per-region $\text{gdp_per_capita} + \text{unemployment_rate} + \text{EP_rate_region} + \text{elderly_pct} + \text{computed V_socio}$ (via mirrored $0.45 \cdot \text{ep_norm} + 0.35 \cdot \text{gdp_norm} + 0.20 \cdot \text{elderly_norm}$ formula from socioeconomic.py:511–517 canonical). Task #453 is the ADMIN structural variant within REPORTS_FRAMING_KB.md §8bis Discipline #47 spatial-enrichment family (STOCK = Task #451 GHSL zonal-sum, FLOW = Task #452 Niva point-sampling, ADMIN = Task #453 polygon spatial-join). Discovery motivating Task #453 also codified METHODOLOGY_DISCIPLINES.md §5septies (empirical OSM tag density is per-country) as candidate architectural discipline — Poland + Luxembourg + Slovenia + Colombia + Lithuania empirically ALL show 0.0% populated ref:nuts:3 / addr:department tag rates at country scale, retiring the connector-tag-extraction path as load-bearing for admin-code derivation. Task #454 SYSTEMIC (24 July 2026) extended the Task #453 polygon utility to 11 additional countries with matching v43-at-partial-coverage pattern — 9 EU countries (Belgium + Czechia + Denmark + Estonia + Finland + Ireland + Latvia + Netherlands + Poland) via the same Eurostat GISCO NUTS-3 2024 shapefile already loaded for Task #453, plus 2 non-EU countries via GADM 4.1 admin1 shapefiles (Switzerland cantons + Canada provinces; same UC Davis academic-open-license documented-proxy pattern used for Task #453 Colombia). Two utility refinements landed during Task #454 execution to accommodate new failure modes surfaced empirically: (a) a 3-case skip-logic decision tree with new case-b (province populated by earlier connector pass but socio_economic gap unfilled — Canada case; utility bypasses polygon join and uses the existing province directly as CSV lookup key); (b) Switzerland csv_lookup_aliases with 5 entries mapping GADM English/French canton names (Lucerne, Sankt Gallen etc.) to CSV German canonical forms (Luzern, St. Gallen). Task #454 SYSTEMIC empirical closure: 52,043 v43 substations enriched across 11 new countries (BE 5,428 + CZ 7,825 + DK 2,364 + EE 1,178 + FI 135 + IE 282 + LV 3,425 + NL 3,807 + PL 25,509 + CH 865 + CA 1,227) plus Task #453 4-country idempotent no-op preservation of 6,967 subs = 59,041/59,077 = 99.94% cohort-wide coverage across 15-country combined Task #453+#454 cohort with 65 Convention #56 out-of-polygon fallback (0.11%) and 100% Task #451/#452 marker preservation via merge-not-replace BINDING contract. Wall-clock 15s cumulative (Poland alone 1.3s = ~19,600 subs/sec). Post-Task-#454 Discipline #47 ADMIN variant instance count is 15 (Task #453 4 + Task

#454 11), and Convention #78 §5 septies OSM-tag-density empirical instance count is 16 (Poland P21 baseline + Task #453 cohort 4 + Task #454 cohort 11 — well above Convention #76 BINDING promotion threshold of 5–10 empirical instances). All inputs sourced from institutional statistical offices with citable vintage — meeting CSRD Article 29a limited assurance requirements for the 32-country cohort where fields are per-substation, plus Task #453+#454 15-country cohort now closed at Grid Equity axis via polygon spatial-join.

3 Infrastructure Resilience [Re composite home]

FC v3 §14 · SSI v4.2 Re composite (canonical) · EU Taxonomy Art. 11 (mapped)

READY

WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate change adaptation in infrastructure. This is the lowest-hanging ESG product from the SSI Index — READY for 9 of 10 countries today. The SSI composite score (R_median), 6 components, modifier architecture, and Markov risk outputs provide the quantitative evidence base for Article 11 screening at asset level.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. By classifying this substation against Article 11 technical screening criteria — using R_median (0.895), all 6 components, and 5 modifiers — this report provides asset-level Taxonomy alignment that enables sustainable finance reporting.

SDG 9

SDG 13

SDG 12

Substation Profile

R_median (composite)	0.895
R_base (pre-modifier)	0.367
Modifier Impact	143.6%%
C (Condition)	0.404
V (Voltage)	0.278
I (Insulation/Climate)	0.355
E (Environment)	0.300
S (Seismic)	0.410
T (Transition)	0.352

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R_median Composite score	0.895	EU Taxonomy Art. 11: Adaptation screening	READY
6 Comp. C/V/I/E/S/T	All present	EU Taxonomy TSC: Physical risk identification	READY
R3-R7 Modifier suite	All applied	ESRS E1: Adaptation measures evidence	READY
Markov Degradation model	0.207	TCFD: Forward-looking risk assessment	READY
R5 Asymmetric CI	0.240	ESRS: Uncertainty quantification	READY

Data Sources & Vintage

ERA5 Climate Reanalysis
Copernicus CDS

2024 Weekly

National Seismic Hazard Map
National Geological Survey

2023 Multi-year

Economic Statistics (GDP, unemployment)
National Statistics Office

2023 Annual

CIGRE TB 761 Markov
CIGRE

2019 N/A

TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6-component weighted composite ($C=0.30$, $V=0.10$, $I=0.25$, $E=0.10$, $S=0.20$, $T=0.05$), Gaussian copula Monte Carlo (10,000 iterations), and 5 modifiers (R3 social, R4 topology, R5 asymmetric CI, R6a restoration, R7 cyber). The Markov 5-state degradation model provides the forward-looking element. All inputs are from institutional open-source data with citable vintage — meeting CSRD Article 29a limited assurance requirements.

4

Pollution & Corrosion

ESRS E2 Pollution · ISO 9223 Corrosion Classification · Environmental Impact

READY

WHY THIS REPORT EXISTS

Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. ESRS E2 requires disclosure of pollution risks. This report assesses the substation's corrosion classification under ISO 9223 and the environmental sensitivity of its surroundings.

SDG Alignment

SDG 11 — Sustainable Cities and Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The corrosion class (C3) and E2_local enrichment factor (1.02) track environmental degradation and pollution sensitivity at asset level. Where corrosion is advanced and maintenance deferred, equipment failure risks releasing mineral oil, PCBs (legacy units), and SF6 — directly relevant to urban and peri-urban environmental quality.

SDG 11

SDG 3

SDG 15

Substation Profile

Corrosion Class (ISO 9223)	C3
E2 Local Enrichment	1.02
E Component Score	0.300
Markov Steady-State	44%/29%/18%/9%

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
Corrosion ISO 9223 class	C3	ESRS E2: Pollution risk classification	READY
E2_local Environmental sensitivity	1.02	ESRS E2: Local environmental impact	READY
E Environment component	0.300	ISO 9223: Atmospheric corrosion	READY

Data Sources & Vintage

Energy Market Data National TSO	2023 Annual
Renewable Installations National DER Registry	2024 Monthly

TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Status depends on whether the corrosion class shows real variance across the fleet or uses default values. Full validation requires national environmental agency air quality monitoring overlay at substation coordinates (SO2, NOx, particulate deposition).

5

Energy Transition & DER Stress

ESRS E1 Transition Plan · TCFD Transition Risk · EU Green Deal Alignment

READY

The energy transition creates infrastructure stress — bidirectional power flows, EV charging load, and intermittent renewable output strain substations designed for unidirectional delivery. This report measures whether grid infrastructure is keeping pace with decarbonisation.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.2 requires substantially increasing the share of renewable energy. This substation has a DER ratio of 0.598. The T1_score (0.534) measures the stress this places on the substation, providing the empirical feedback loop between SDG 7 ambition and infrastructure reality.

SDG 7

SDG 13

SDG 9

Substation Profile

T1 Score (transition stress)	0.534
DER Ratio	0.598
DER Variability	0.654
EV Load Ratio	0.087
T Component Weight	0.05

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
T1 Transition stress score	0.534	ESRS E1: Transition plan assessment	READY
DER_ratio Renewable/load ratio	0.598	TCFD Transition: Technology risk	READY
DER_var Intermittency	0.654	EU Green Deal: Grid flexibility	READY
EV_load EV charging ratio	0.087	SDG 7: Clean energy access	READY

Data Sources & Vintage

Air Quality Monitoring
National Environmental Agency

2023 Annual

ISO 9223 Corrosion
ISO

2012 N/A

TECHNICAL VALIDITY

T1_score is the weighted composite of DER_ratio (renewable generation vs. local demand), DER_variability (intermittency), and EV_load_ratio (electric vehicle charging load as fraction of capacity). DER data sourced from national energy regulators and renewable installation registers. DER_ratio > 1.0 indicates net export during peak generation — a marker of successful energy transition with associated infrastructure stress.

6 Cybersecurity Exposure

NIS2 Directive · ENISA Cybersecurity Index · ESRS G1 Governance

PARTIAL

WHY THIS REPORT EXISTS

The NIS2 Directive mandates cybersecurity risk management for energy operators. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment, communication architecture analysis, and national-level cyber maturity (ENISA/DESI indices) to produce a substation-level digital vulnerability score.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems, this must include cyber resilience. The R7 modifier (1.029) assesses digital vulnerability combining SCADA exposure, communication protocol security, and national cyber maturity. SDG 16 (Strong Institutions) also applies: NIS2 compliance is a governance imperative, and the betweenness centrality (0.333) identifies single-point-of-failure risk in the grid topology.

SDG 9

SDG 16

Substation Profile

R7 Cyber Modifier	1.029
Graph Degree (topology)	5
BC Percentile (centrality)	0.333
Is Bridge Node	No
Cluster Coefficient	0.0186

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R7 Cyber modifier	1.029	NIS2: Cybersecurity risk management	READY
BC Betweenness centrality	0.333	ESRS G1: Governance & topology risk	READY
Degree Graph degree	5	Grid resilience: Connectivity	READY

Data Sources & Vintage

Cybersecurity Index ENISA	2024 Biennial
DESI Connectivity European Commission	2024 Annual

TECHNICAL VALIDITY

R7_cyber is computed from a weighted combination of: (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators, (2) graph topology vulnerability (betweenness centrality = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. Substation-level SCADA data is operator-proprietary — the national-level proxy provides a baseline but not asset-specific granularity.

WHY THIS REPORT EXISTS

The SFDR (Sustainable Finance Disclosure Regulation) mandates that financial market participants disclose Principal Adverse Impact (PAI) indicators for infrastructure investments. PAI Table 1 requires quantitative assessment of physical + governance risk at asset level for Article 8/9 fund classification. The SSI v4.2 Re composite (resilience metric per FC v3 §14 subsection 13.7) provides the canonical proxy for Infrastructure PAI: it integrates 6-axis physical risk (C/V/I/E/S/T), 6-modifier degradation chain (R3/R4/R6c/R6d/R6e/R8/R9/R10), and Markov forward-looking trajectory into a single normalised value directly consumable by SFDR reporting.

SDG Alignment

SDG 12 — Responsible Consumption and Production (primary)

Target 12.6 encourages companies to adopt sustainable practices and integrate sustainability information into their reporting cycle. SFDR is the EU regulatory instrument that operationalises this at fund level for infrastructure investors. This substation carries an Re_norm (normalised resilience composite) of 1.000 and Re_raw of 1.787, providing the empirical basis for PAI Table 1 Infrastructure indicator disclosure under Delegated Reg (EU) 2022/1288.

SDG 12

SDG 9

SDG 13

Substation Profile

Re_norm (SFDR PAI composite)	1.000
Re_raw (pre-normalisation)	1.787
R6c Flood modifier	1.300
R6d Wildfire modifier	1.190
R6e Winter storm modifier	1.131
R8 Adaptation modifier	0.932
R9 Compound event modifier	1.100
R10 Just-transition modifier	1.120

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
Re_norm SFDR PAI composite (normalised)	1.000	SFDR Art. 4 · PAI Table 1 Infrastructure	READY
Re_raw Pre-normalisation resilience	1.787	FC v3 §14 subsection 13.7	READY
R6c Flood modifier	1.300	SFDR PAI: Physical acute hazards	READY
R6d Wildfire modifier	1.190	SFDR PAI: Physical acute hazards	READY
R6e Winter storm modifier	1.131	SFDR PAI: Physical acute hazards	READY
R8 Adaptation modifier	0.932	SFDR PAI: Adaptation planning	READY
R9 Compound event modifier	1.100	SFDR PAI: Compound climate risk	READY
R10 Just-transition modifier	1.120	SFDR PAI: Social transition risk	READY

Data Sources & Vintage

No dedicated data sources for this report.

TECHNICAL VALIDITY

Re_norm is the canonical SFDR PAI Infrastructure composite per SSI Index v4.2 methodology (FC v3 §14 subsection 13.7). It integrates the 8 v4.2 modifiers (R6c/R6d/R6e physical acute hazards + R8 adaptation + R9 compound events + R10 just-transition + R7 cyber + R3 social) into a single value normalised against the country-fleet percentile distribution. Convention #56 visibly-honest degradation preserves Re_norm=0.0 + Re_raw=1.0 as neutral defaults for net-new substations awaiting the L2/L3/L4 modifier-chain rescore pass per Convention #78 §4bis.4 two-phase workflow. GAP status reflects sites with fewer than 50% of fleet subs carrying non-default modifiers (typical during a v4.23 L1 refresh window before the follow-on rescore).

SFDR Principal Adverse Impact (PAI) Infrastructure v4.2 — Re

7

composite anchor

SFDR Article 7 PAI · ESRS E1-7 climate-resilience disclosure · CSRD double-materiality infrastructure overlay (per FC v3 §14 subsection 13.7)

READY

SFDR Article 7 requires asset-level disclosure of Principal Adverse Impacts on sustainability factors for infrastructure investments. The Re composite (v4.2 bounded resilience aggregate, bounds [0.920, 1.787] normalised to [0, 1]) is the SSI's canonical driver for PAI Infrastructure exposure — it captures the substation's standing on the new v4.2 resilience-modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in a single scalar suitable for SFDR PAI reporting. CSRD double-materiality requires both inside-out (impact on environment) and outside-in (climate-resilience risk to the asset) — Re composite anchors the outside-in dimension for grid infrastructure. This report closes the FC v3 §14 subsection 13.7 R7 SFDR PAI axis on the ESG Radar.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for developing quality, reliable, sustainable and resilient infrastructure. The Re composite $Re_norm = 1.000$ (elevated exposure) at this substation provides the SFDR PAI Infrastructure metric, anchoring climate-resilience disclosure to a single auditable scalar. $Re_raw = 1.7870$ with methodological bounds [0.920, 1.787] derived from the v4.2 modifier family product range. Lower $Re_norm \rightarrow$ lower PAI Infrastructure exposure $\rightarrow$ stronger SFDR Article 7 alignment.

SDG 9

SDG 13

SDG 11

Substation Profile (Re composite)

Substation ID	120000037277
Re_raw (unbounded)	1.7870
Re_norm (bounded [0, 1])	1.000
Tier classification	ELEVATED
Methodology bounds	[0.920, 1.787]
United States fleet observed range	1.155 $\rightarrow$ 1.200
Coverage	45,003 substations

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R6c Flood (ISPRA PGRA, ADDITIVE +0.00..+0.30)	+1.300	SFDR PAI 4 (climate physical risk)	READY
R6d Wildfire (EFFIS + FIRMS + SIAB, MULT)	1.190	SFDR PAI 4 (climate physical risk)	READY
R6e Winter / Ice (ERA5 Bourgouin p99, MULT)	1.131	SFDR PAI 4 (climate physical risk)	READY
R8 Adaptive Capacity (FERC + EIA + NARUC)	0.932	ESRS E1-3 Adaptation actions	READY
R9 Compound Hazard (ISPRA RNDA, STEP)	1.100	ESRS E1-9 Anticipated financial effects	READY
R10 Energy Justice (US Census + EIA RECS)	1.120	SFDR PAI 14 (board diversity proxy) + ESRS S3	READY
Re_norm aggregate (bounds [0.920, 1.787])	1.000	SFDR Art. 7 PAI Infrastructure anchor	READY

Data Sources & Vintage

ISPRA IdroGEO PGRA	2024 (annual)
Copernicus EFFIS + NASA FIRMS	Q4 2025 (active + historical)
ISPRA SIAB national-tier burned-area	CY2024
ECMWF ERA5 Bourgouin freezing-rain p99	1991-2020 reanalysis
FERC + EIA + state PUC + 3,000+ utilities smart-meter penetration	2024 annual
ENEA Adaptation Index	2024
DESI Digital Economy index	2024
ISPRA RNDA compound-hazard cross-product	CY2024
EU-SILC ILC_MDES01/03	2024 wave
US Census ACS + EIA RECS + LIHEAP	2024
EIA-861 regional SAIDI (Annual Electric Power Industry Report)	CY2024

TECHNICAL VALIDITY

The Re composite is the bounded aggregate $Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$ with normalisation $Re_norm = clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1)$. Bounds [0.920, 1.787] derived from the multiplicative band of the 5 multiplicative v4.2 modifiers at tier extremes plus the R6c additive cliff range. United States fleet observed range [0.974, 1.604] across 45,003 substations. SFDR PAI Infrastructure proxy validated against TCFD physical-risk framework and ESRS E1-9 anticipated-financial-effects disclosure. See FC v3 §14 subsection 13.7 for the R7 SFDR PAI axis derivation.

ITEM	VALUE
Scoring Engine	SSI Index v4.2 — Gaussian copula Monte Carlo (10,000 iterations) · 11 modifiers (5 baseline + 6 v4.2 resilience family) · Re composite bounded [0.920, 1.787]
Weight Architecture	C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05
Modifiers Applied	R6c flood: 1.300, R6d wildfire: 1.190, R6e winter: 1.131, R8 adapt: 0.932, R9 compound: 1.100, R10 just: 1.120, R6 seismic: 1.010, R3 C_mult: 0.973, R7 cyber: 1.029, R4 F_topo: 0.998, R6 restoration: 1.038
Degradation Model	Markov 5-state, CIGRE TB 761 calibration
Thermal Model	IEEE C57.91 transformer loading curves
Report Date	17 March 2026
Reporting Period	Calendar year 2025
Refresh Cadence	Annual (data ingestion → scoring engine → display)
Substation	Fermilab Kautz Road Substation (120000037277)
Data Assurance Level	All inputs from institutional open-source with citable vintage
Re composite (v4.2)	Re_raw = 1.7870 · Re_norm = 1.000 · ELEVATED exposure · bounds [0.920, 1.787]

D Re composite — bounded resilience aggregate (v4.2)

Resilience composite anchoring CSRD/ESRS climate-resilience disclosure

The Re composite (v4.2) is a bounded resilience aggregate capturing the new R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just modifier family in a single scalar suitable for CSRD/ESRS climate-resilience disclosure:

$$\text{Re_raw} = (\text{R6d} \times \text{R6e} \times \text{R8} \times \text{R9} \times \text{R10}) + (\text{R6c} - 1.00)$$

Bounds: **[0.920, 1.787]**. Re_raw < 1.000 when R8_adapt < 1.00 dominates other near-neutral factors. American fleet observed range across 101,594 substations: the methodology-bounded extremes [0.920, 1.787] — low-end values arise at interior low-exposure substations with R8_adapt < 1.00 dominating near-neutral other modifiers, high-end values at coastal compound-exposure substations where R6c flood + R9 compound stack multiplicatively. v4.2 explicitly retires the pre-Path-B "winter exposure is a real climate burden" reading per Convention #6 §2.3.1 (Bourgouin freezing-rain p99 tail-event amplifier per §6 design — winter exposure signal correctly lives in v4.0.2 II Snow/Ice metric, where it methodologically belongs). See [W1-W10 FC v2](#) for per-substation worked examples + W4 Climate resilience baselines.

SSI Index v4.0.2 · Ikenga Enhanced Infrastructure Intelligence · Annual ESG Disclosure

Report generated 21 May 2026 · All data sources are open-access institutional

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.